Image-based Object Insertion using Persistent and Transient Decomposition

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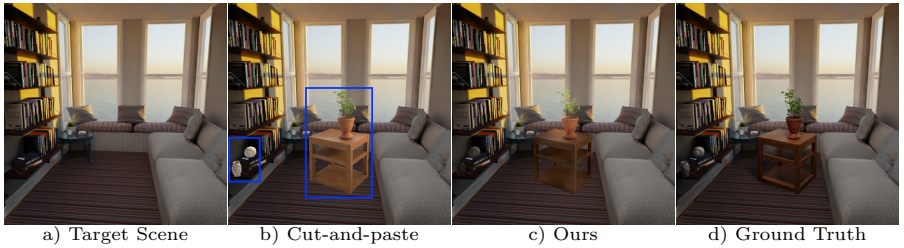

a) Target Scene b) Cut-and-paste c) Ours d) Ground Truth

Fig. 1. Inserting objects from one image into another requires estimating the illumination of the target scene and re-lighting the objects using their geometry. Otherwise inserted objects will look out of place (cut-and-paste). We present an image-based approach to object insertion, which produces convincing compositions, plausibly matching the environment lighting (ours) using a self-supervised setup.

Abstract. We describe Illuminator, a self-supervised method that can convincingly insert objects from image information alone. We design Illuminator to compose images from intermediate image representations that disentangle images into persistent and transient maps. Our design forces the Illuminator to implicitly infer the illumination field of a scene. To improve generalization to a wide variety of scenes, we introduce a novel cross-illumination cycle consistency loss that enables the abstraction of illumination of the scene from transient maps. Training requires only in-the-wild images (rather than costly paired or CGI data). For quantitative evaluation, we introduce a synthetic dataset of objects inserted into multiple scenes. We compare our method against several challenging baselines that represent the state-of-the-art for different perspectives on the problem and show significant improvements over all baseline methods both qualitatively and quantitatively.

1 Introduction

Insertion rendering cuts **fragments** (typically, object regions) out of a **source image** and places them into a **target image** (a in Figure 1) to produce a **cut-and-paste image** (b). This is easy to do, but often results in unnatural images (as in Figure 1), and we describe a method to correct the result into a **rendered image** (c). Reliably successful insertion rendering would be extremely useful. For example, we could create easy scene editing applications for unskilled users, or very rich training data resources [10, 18]. Modeling and rendering

complex material properties in computer graphics images (CGI) remains very difficult; successful insertion rendering would preserve these properties just by using pictures of real objects. The key obstacle is shading: the illumination field in the source image may differ greatly from that in the target image, meaning that straightforward cut-and-paste results in jarring inconsistencies. When object geometry and material are known, one can estimate illumination in the target image and shade using that [17, 31]. It is not currently practical to use these methods to improve insertion rendering because it is extremely hard to recover 3D object geometry and material from a single image at a sufficient level of quality (say, a lego dozer or furry cat) [4, 32]. This paper describes a completely self-supervised method – Illuminator – that can convincingly transfer objects from image to image. The method is learned using a standard in-the-wild dataset [35], without using geometric data, environment maps or CGI.

For a successful insertion, Illuminator must distinguish between **persistent effects** and **transient effects**. Persistent effects should not change when an object moves from scene to scene. Transient effects may change depending on the background scene's lighting and layout. We investigate the use of standard decompositions into albedo and shading, and find poor results over a number of strong intrinsic image methods (Section 6). The cause appears to be a tendency for albedo maps to omit important details of appearance in high spatial frequencies.

Our results are sharply improved by using a modified decomposition. We use a learned image decomposition procedure, trained with samples from spatial models as in [14], to obtain a decomposition that preserves these details. We decompose image into three components: a persistent map, a multiplicative transient map and an additive transient map. Our persistent map can be compared to albedo but it also encodes high-frequency shading information. Our multiplicative transient map can be compared to diffuse shading and our additive transient map can be compared to glossy or specular effects in an image. However, it is still quite challenging to get an ideal decomposition – image referred maps of diffuse and specular effects (transient map) may still contain persistent information. For example, information about surface normals. To this end, we train Illuminator to produce an abstracted internal representation of transient illumination conditions with a novel cross-illumination cycle consistency loss (Section 5).

Quantitative evaluation is by direct comparison – we transfer an object from a CGI scene to another CGI scene, then compare to a CGI image containing the object. We introduce a new synthetic dataset of objects inserted into multiple scenes for image-based object insertion evaluation. We compare a number of recent methods, including current SOTA for: different decomposition methods [14,30,31], image harmonization [8,36]; and CGI object insertion [31] (which gives a form of an upper baseline on performance). Our evaluation shows that: (a) Illuminator can produce insertion rendered images with higher PSNR than any comparable method; (b) the choice of image decomposition is crucial; (c) the traditional decomposition metric (WHDR) is a poor predictor of insertion rendering success; (d) the illumination cross-loss results in important improvements to performance.

Qualitatively, Illuminator performs well at a variety of tasks, including insertion rendering, portrait relighting and scene relighting.

In summary, our main contributions are: (1) the first self-supervised, image-based object relighting method trained without labeled paired data, CGI data, geometry, or environment maps. (2) A novel cross-illumination cycle consistency loss that enables abstraction of the scene's lighting representation. (3) A new class of image decomposition into persistent and transient maps designed on the pragmatics of what should change and what should not. (4) A rendered dataset of 12 different objects under different lighting conditions for evaluating and bench-marking future insertion methods. (5) Finally, a method quantitatively outperforming all reasonable state-of-the-art (SOTA) comparisons.

2 Related work

Image-based object insertion. Poisson blending [23,41] based object insertions often lead to color bleeding from the background into the inserted object resulting in color changes, strong boundary artifacts, and displeasing results. [28,52] extend object insertion to a data-driven approach by retrieving objects that match target scene lighting, but cannot correct lighting – so, for example, a fragment cut from a bright image and placed in a dark image will seem to glow. In contrast, our method automatically adjusts lighting to produce an image that is like real images. Semantic context-based editing [2, 29] uses scene context and labeled annotations. Bhattad and Forsyth [5] adjust the shading field using a deep image prior [51] with surface normal supervision. In contrast, our method requires no annotation. Image harmonization (IH) methods [8, 36, 49, 50] could be used to produce a rendered image, but tend to adjust both background and object. In contrast, by respecting the distinction between persistent and transient effects, our method can correct a fragment without misrepresenting the background.

Computer graphics (CG) object insertion is well established. Karsch et al., [25,26] and Kholgade et al., [27] show convincing insertion of CGI objects into photographs. Recently, Chen et al., [7] insert cars from a collection of 3D CAD models into scenes. Learning to recover scene lighting from images is now highly developed (eg [15–17, 31, 46, 47, 53]), supporting strong rendering of CGI objects. CGI insertion methods should be better than fragment insertion methods, and we compare to [31] as an upper baseline. But image fragments do not come with geometry or material models, and accurately recovering such models in sufficient detail from single images remains hard [4, 19, 32]. One strategy to insert a fragment is to impute a crude geometric model of the object, and work with that [33, 34]. In contrast, we exploit models of what does and does not change when objects appear on different backgrounds from in-the-wild data.

Image-based Relighting. Recent works show portrait images can be convincingly relighted [40, 44, 48, 56]; these methods are specialized to faces, and are trained using paired or light-stage data. In contrast, we show acceptable portrait and scene relighting results from a method trained entirely on scene images.

Decomposition of an image into persistent and transient effects is key to our method, and is rooted in intrinsic image decomposition [6,12,22,30]. Rather than decompose into albedo and shading, as is usual, we distinguish between what should not change and what should; but we use procedures from the intrinsic image literature. Forsyth and Rock [14] uses authored examples (paradigms) to train an intrinsic image decomposition method with strong quantitative performance; we adopt that method to identify persistent (resp. transient) effects *by example*. As in Section 7, we author examples of persistent effects (like the Mondrian of the traditional intrinsic image literature, review in [14]); multiplicative transients (like the shading field in [14]); and additive transients (white blobs, to model gloss). Intrinsic images are currently evaluated using the weighted human disagreement rate (WHDR) [3]. While our persistent maps are analogous with albedo (and have reasonable WHDR), we find WHDR does not predict the results of direct evaluation of insertion rendering, suggesting that the distinction between albedo (a physical quantity) and persistent effect (a phenomenological idea) is important.

Cycle Consistency. Cycle consistency losses has been used in various computer vision tasks [1,4,38,39,57,58]. Most of these works deploy cycle consistency to improve the perceptual quality of rendered images. We use cycle consistency to extract illumination representation from their transient maps.

3 Approach

A user cuts a fragment I_{obj} from a source image I_{src} and places it into a target image I_{dst}, forming a cut-and-paste image $I_{c\&p}$ which is often not realistic. We will build a network, Illuminator f which accepts as **input** this cut and paste image and produces a **corrected image** I_c. More formally, $I_c = f(I_{c\&p})$, where the cut-and-paste image is composed by blending the fragment image with the background: $I_{c\&p} = M_{obj} \circ (I_{obj}) + (1 - M_{obj}) \circ I_{dst}$. M_{obj} is a blending mask.

At inference, we generate the corrected image from our network in three key steps: **decompose** the input into a map of persistent and transient effects; **abstract** the transient effects of the target scene into a form that can be applied to any arbitrary persistent map; finally, **decode** a new (corrected) image out of the persistent map and the abstraction. Figure 2 shows our network. We use a simple U-Net with two encoders: one for persistent map and another for the transient map.

Decomposition. For training such a network, we require an image decomposition dataset. However, we wish to train Illuminator without relying on any external labeled dataset like CGI. We assume for now we have a reliable source (a pretrained network) that decomposes images into two maps – a persistent map (P) and a transient map (T). We form an image as $I = P \times T$. For comparison, we also study traditional image decomposition – albedo as persistent map and shading as transient map – from three different SOTA methods [14,30,31] in Section 6. We discuss the role and type of image decomposition crucial for Illuminator later in detail in Section 7.

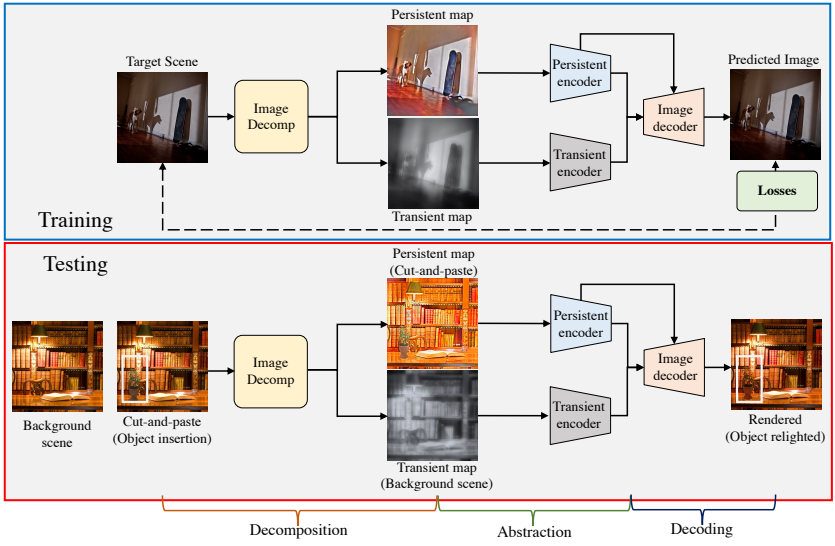

Fig. 2. Method overview: Illuminator corrects a cut-and-paste input into a realistic image by decomposing, abstracting and decoding. We decompose images into a persistent map (does not change when moving from scene to scene) and a transient map (that may change) and encode them into persistent and transient codes. We decoded these codes into the corrected image. Our training uses a novel cross-illumination consistency loss (Fig. 3) to force the transient code to abstract scene illumination; no marked-up data is required and we train Illuminator with in-the-wild images. At test time, Illuminator sees the persistent map for the cut-and-paste, and the transient map for the target scene, and produces a corrected version of the input.

Abstraction. It is essential to abstract. The persistent map contains information that is not changed when objects move from image to image. This implies persistent map for the corrected image should be the same as that for the input. But the transient maps for the inserted fragment are wrong; otherwise, there would be nothing to correct. We must recover information about lighting from the transient maps of the target. It is not enough to just use these maps directly, because they encode information on the shape (e.g. normals) of the target scene that should not be applied directly to the fragment. Instead, the network must recover some abstraction of lighting from the transient maps. Rather than using some parametric representation (cf spherical harmonics), we use a novel loss to force our network to abstract. This means we do not require labeled illumination data. We discuss our abstraction procedure in Section 5.

Decoding. In principle, we could require the network to construct transient maps for the corrected image, then recompose with the persistent map. But this means that any error in the decomposition is preserved and will reflect in our corrected image. Instead, we will pass encoding of the persistent map and encoding of abstraction to a decoder. We force the decoder to produce the results we want by using several consistency losses. We elaborate on this step next.

4 Decoding

We force Illuminator to decode these latent codes from persistent and transient maps into a real image by imposing simple consistency losses. Note at training, Illuminator sees only real images rather than cut-and-paste because we do not have ground truth for the corrected version of cut-and-paste. We use MSCOCO [35] because: it is widely known; it has a diverse collection of indoor and outdoor scenes with a wide variety of objects and lighting scenarios; and we require no special annotation. We assume the object represented by the fragment does not significantly affect the scene illumination (so we can't, say, put covers over lights). This means that correction should produce a result whose background is a lot like the target background, but where the fragment is consistently shaded.

Target consistency: In training, the corrected image I_C should be the original image I_O, so we impose an $\mathcal{L}_1$ reconstruction loss

$$\mathcal{L}_{rec} = ||I_O - I_C||_1. \tag{1}$$

We use a perceptual loss [24]from a VGG [45] feature extractor (Φ) at various feature layers (j) to control texture errors

$$\mathcal{L}_{per} = ||\Phi_j(I_O) - \Phi_j(I_C)||_2. \tag{2}$$

Decomposition consistency: We want persistent maps for the corrected image to be close to that of the original, and we want transient maps to be similar, too. Our image decomposition consistency loss is given by

$$\mathcal{L}_{decom} = ||P(I_O) - P(I_C)||_2 + ||T(I_O) - T(I_C)||_2 \tag{3}$$

Realism: We want corrected images to be realistic, and so we use an adversarial loss (non-saturating). Our generator G is Illuminator with parameters ψ and Discriminator D with parameters θ. For our discriminator, we find a vision transformer [9,20] based discriminator to provide better object relighting when comparing to convolutional discriminator. We provide an ablation study for it in our supplementary. So our adversarial losses are

$$\mathcal{L}_{\text{adv}} = f(D_\psi(G_\theta(I_C))) + f(-D_\theta(I_O))\text{where } f(u) \triangleq -\log(1+\exp(-u)). \tag{4}$$

Our base losses are

$$\mathcal{L}_{base} = \lambda_{rec}\mathcal{L}_{rec} + \lambda_{per}\mathcal{L}_{per} + \lambda_{decom}\mathcal{L}_{decom} + \lambda_{adv}\mathcal{L}_{adv} \tag{5}$$

The corresponding λ for each loss term are in our supplementary.

5 Abstraction from Cross-Illumination Cycle Consistency

Shading at a point is determined by both the normal at that point and the surface irradiance. For example, the shading field of a cluttered room and the same room

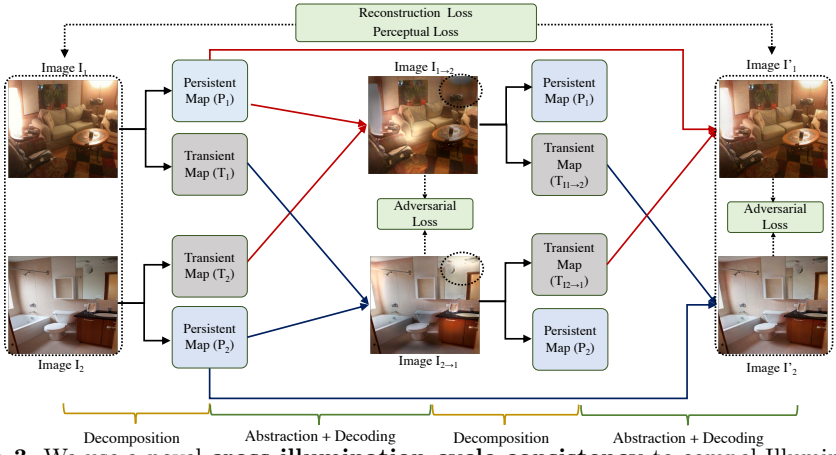

Fig. 3. We use a novel **cross illumination cycle consistency** to compel Illuminator to disentangle the transient map into an abstract illumination representation. In training, we swap the transient maps of randomly chosen pairs of images; this results in intermediate relighted images. The intermediate images are forced to be realistic by an adversarial loss, and so the geometric information in the transient map (normals in the shading) must be suppressed. We then swap back the transient maps for these intermediate relighted images and require that Illuminator reproduce the originals. This forces Illuminator to recover only scene irradiance from the transient maps. Notice how the light source in the top scene is removed (**upper right corner**) and added to the bottom scene (at the same location) in our intermediate images.

without clutter might be wildly different, even if a sconce on the wall illuminates both. We want to transfer some model of the distribution of irradiance, but not surface normals. This means we want to transfer the presence of the sconce without the normals. Parametric models of illumination come with disadvantages. For instance, spherical harmonic illumination means that points with the same normal have the same shading. Therefore, we do not want to impose a model of illumination. Instead, the method should choose its own representation of illumination so that it can opportunistically represent the effects it observes in data. We achieve this with a novel **cross illumination cycle consistency loss**. For brevity, we henceforth refer to this loss as cross-loss. This means we do not require labeled illumination data.

The cross-loss exploits the following idea. Assume we relight image I_1 using some other image I_2's transient maps to get $I_{1\to 2}$; then decompose $I_{1\to 2}$; and then construct I_1' from $I_{1\to 2}$'s persistent map and I_1's transient maps). Persistent maps don't change when lighting changes and transient maps do. This means that there should be no difference between I_1' and I_1 if the method uses only illumination information extracted from transient maps. Given two images, I_1 and I_2, we obtain persistent maps (P_1, P_2) and transient maps (T_1 and T_2. We then swap their transient map to obtain, using Illuminator R:

$$I_{1\to 2} = \mathrm{R}(P_1, T_2)$$
$$I_{2\to 1} = \mathrm{R}(P_2, T_1)$$

Table 1. Current SOTA image decomposition methods produce poor insertion rendering. IR++ is when using *predicted* environment maps from [31] and ground truth 3D geometry to render insertions using a physics-based engine.

Method	Self-Sup	PSNR↑	MSE↓	LPIPS↓
IR++ [31]	✗✗	18.84	1574.84	0.033
Naive Cut-and-Paste		15.08	3648.84	0.042
CGIntrinsics [30]	✗	15.68	3211.58	0.046
IR [31]	✗	14.60	4435.18	0.050
Paradigms [14]	✓	15.20	3127.21	0.046

We now infer persistent and transient from these two new images and swap transient maps back again to get the original images:

$$I_1^{'} = \mathrm{R}(P_1, T_{MI_{2\to1}}, T_{AI_{2\to1}})$$
$$I_2^{'} = \mathrm{R}(P_2, T_{MI_{1\to2}}, T_{AI_{1\to2}})$$

Our loss now compels I_1 and $I_1^{'}$ (etc.) to be similar (see Figure 3).

Our final losses to train illuminator also includes these additional consistency that we get from cross-loss ($\mathcal{L}_{cross}$). Therefore, our final losses are

$$\mathcal{L}_{final} = \mathcal{L}_{base} + \lambda_{cross}\mathcal{L}_{cross} \tag{6}$$

6 Experiments using Traditional Image Decomposition

We first evaluate Illuminator's performance when using traditional image decompositions – albedo as persistent map and shading as transient map. We compare two supervised SOTA methods [30,31] and one self-supervised SOTA method [14]. We used pretrained models provided by these papers to get decomposition maps.

Synthetically rendered dataset. For quantitative evaluation, we generated a synthetic object insertion dataset. We rendered 12 CG objects with complex material and surface properties like a potted plant, ceramic vase, lego dozer, etc., in nine different environments (indoor-outdoor) with two to three different lighting conditions for each. This rendering results in about 4752 quantitative comparisons for object insertion. We provide details in supplementary. Note we will make this evaluation dataset public for future research.

Evaluation. For quantitative comparisons, we compare rendering quality against the ground truth and report PSNR, MSE and LPIPS. We report PSNR and MSE numbers only for rendered fragments and not for the entire image. We show a few qualtiative comparisons in Figure 4 and Figure 5.

Why does traditional image decomposition work badly? From Table 1, results are not strong. 15 PSNR is not great and are comparable to cut-and-paste PSNR. Figure 4 and Figure 5 suggest the cause — the image decompositions we have used do not preserve properties that have important effects on the rendering. For a successful object relighting, Illuminator must distinguish between **persistent effects** (which should not change when an object is moved from scene to scene) and **transient effects** (which may change). This distinction is operational (though rooted in physics) rather than physical. For example, no current method can recover the geometry of a stud on a lego brick from a single

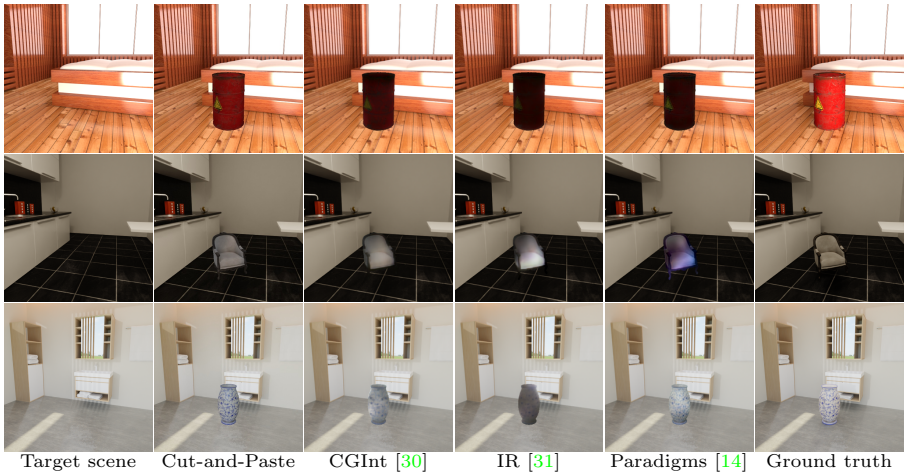

Fig. 4. Poor image-based object insertion renderings because of using current SOTA traditional image decompositions for training illuminator.

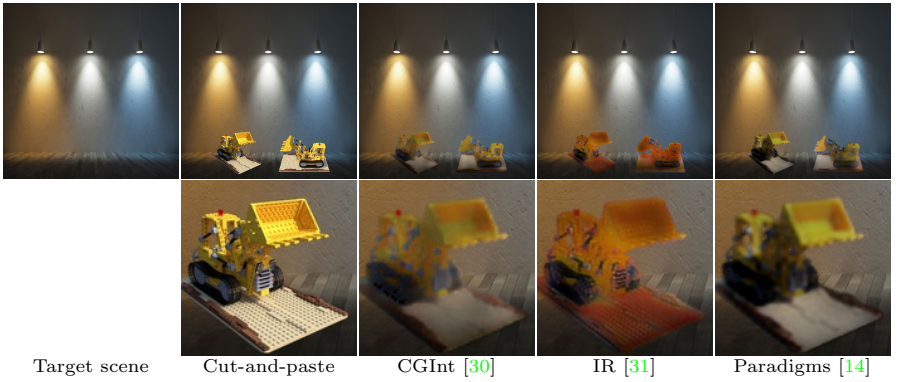

Fig. 5. Current SOTA image decomposition methods do not recover geometric details when relighting objects. **Top:** relightings of a lego object inserted into target scene using various methods. **Bottom:** zoom on the front-end-loader. Cut-and-paste is too bright; other methods blur detail because high-frequency signals of the lego are modeled in the shading field.

image, meaning that the small shadow cast by the stud should be modelled as persistent – because then it will follow the brick when it is moved from image to image. We show one such decomposition in Section 7 (there might be others) that can transfer the lego brick with all the surface details (Figure 6).

7 Improving Results with a Pragmatic Decomposition

Recall that we wish to decompose images into two components – persistent effects that should not change when objects are moved from one image to another image, and transient effects that might – or should – change. Forsyth and Rock [14] train intrinsic image methods using samples from abstract statistical models

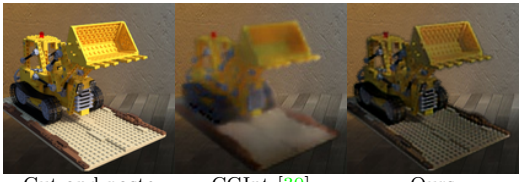

Fig. 6. Our modified decomposition (**right**) preserves surface details like studs on lego. These details may lead to albedo error but are crucial for insertion rendering to work. Images show a crop of an inserted object after correction. See also Figure 5.

(paradigms), where albedo paradigms are Mondrian-like images and shading paradigms are constructed out of Perlin noise [42]. We exploit the fact that their model can be authored, and is extensible. We construct a persistent map and two transient maps. A key difference from our proposed decomposition and traditional decomposition is that we model major high-frequency details in the persistent map and not transient map. However, it is still quite challenging to get an ideal persistent and transient decomposition. This could cause inconsistencies between small and large-scale shadows, but Illuminator encodes then decode architecture driven by a loss that stresses realistic outputs (adversarial losses and cross-loss) means it can interpret persistent maps opportunistically.

We model an image as $P \times T_M + T_A$, where P is a persistent map (analogous to albedo), T_M is a multiplicative transient map (like diffuse shading) and T_A is an additive transient map (like gloss effects). We obtain this decomposition with a learned method described in Forsyth and Rock [14], requiring no labeled CGI data. Hence, our image decomposition network is self-supervised. Finally, training Illuminator using these decompositions makes our entire pipeline self-supervised. We do not use any labeled or CGI data for training. Samples of our persistent and transient maps are in Figure 7. Details of their construction are in supplementary.

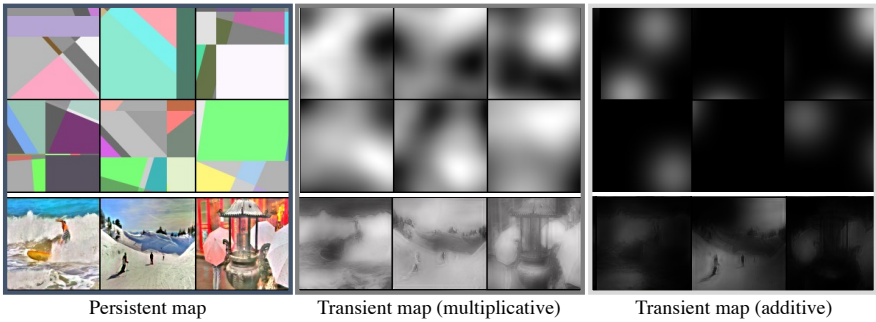

Fig. 7. Top two rows: samples of our persistent and transient maps that we use to train our image decomposition model. Bottom row: decomposition from the randomly chosen MSCOCO images. Unlike traditional albedo and shading decomposition, our decomposition preserves most of the geometric and high-frequency details in the persistent map and hence enable us to transfer objects without loss of surface details.

Table 2. WHDR evaluated on IIW dataset for different image decomposition methods.

CGIntrinsics [30]	IR [31]	Paradigms [14]	Ours
14.80	16.83	16.86	19.00

We empirically show our persistent and transient maps derived from learned spatial paradigm models may rearrange detail as required. They yield convincing object relighting, both qualitatively and quantitatively even though it has worse WHDR scores. Our persistent map achieves WHDR score of 19 and the IIW dataset [3] and SOTA CGIntrinsics [30] achieves WHDR score of 14.8 (see Table 2). Note that this also suggests that decomposition that has strong WHDR may not necessarily support good behavior by Illuminator, because they could omit small details that are important.

8 Experiments using our Pragmatic Decomposition

We now evaluate our proposed decomposition performance for object insertions. Our experimental setup is the same as that is described in our Section 6. When training with our decomposition, we stack multiplicative and additive transient map and feed it to our transient encoder.

Baselines: Image-based object relighting. We compare quantitatively with two SOTA supervised IH baselines [8, 36]. We use naive cut-and-paste as a lower bound for object-relighting. We also provide a strong, self-supervised baseline by simply correcting the inserted object's intensity: we match the mean statistics of our persistent maps from the source object to that of the target scene to account for any changes that may occur because of the changes in the lighting's color from one scene to another. As an upper bound to our method, we render 3D objects (*not* fragments) using environment maps predicted by [31].

Simple intensity correction is a strong baseline. The simple intensity correction baseline improves object-relighting by 7.7% (from 15.08 to 16.24) over cut-and-paste (Table 3) and is better than traditional decomposition (Table 1).

Illuminator beats all baselines. Table 3 shows our self-supervised method improves object relighting performance by 17.04% compared to naive cut-and-

Table 3. Object relighting accuracy when compared to rendered GT. At top, we show an upper baseline to our method that has complete access to 3D geometry of the object. We also show a lower baseline (cut-and-paste). In the middle, we show comparisons to supervised image harmonization baselines [8, 36]. At the bottom, we show comparisons of our Illuminator's insertion using our persistent and transient decomposition. We show a simple intensity correction baseline, surprisingly is very strong. We also report relighting success when compared to IR++. If the PSNR of a rendered object is greater than the IR++'s rendering (upper baseline), we consider it a success.

Method	Self-Sup	PSNR↑	MSE↓	LPIPS↓	Success (%)
IR++ [31] (CGI + 3D Obj Sup)	✗✗	18.84	1574.84	0.033	
Naive Cut-and-Paste		15.08	3648.84	0.042	28.01
Dovenet [8]	✗	16.61	2197.78	0.048	38.15
Rainnet [36]	✗	16.23	2363.16	0.050	35.40
Ours baseline (intensity correction)	✓	16.24	2783.20	0.040	33.06
Ours base model (w/o cross-loss, Eq. 5)	✓	17.16	2134.61	**0.038**	40.24
Ours final model (with cross-loss, Eq. 6)	✓	**17.65**	**2087.03**	**0.038**	**43.52**

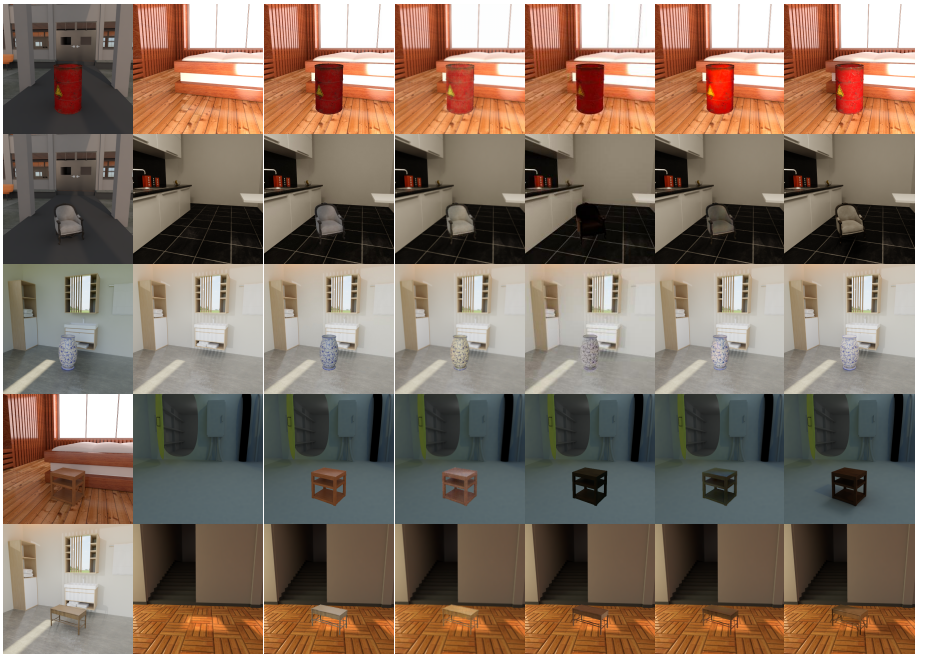

Fig. 8. Comparison on CG rendered dataset. We compare our method with a SOTA IH baseline, a 3D object supervised upper baseline and ground truth images. Our method can infer strong gloss correctly (red barrel in the top tow) that even a 3D supervised method cannot. Illuminator also generates plausible relighted rendering for complex materials like a ceramic vase (middle row). We find IR's environment prediction in scenes with very low luminaire and strong background are inconsistent; thus resulting in incorrect rendering (armchair in the top-middle row).

paste. A SOTA supervised IH baseline [8] improves by only 10.1%. Our cross-loss yields a strong boost (from average PSNR score of 17.16 (13.79%) to 17.65 (17.04%). A qualitative comparison for CG objects rendered is in Figure 8; a qualitative comparison with complex materials in real shaft environments are in in Figure 9.

Normals and depth help, but only slightly. We report an ablation in Table 4 where pretrained normals [11, 54] and depth [11, 43] are used to train Illuminator. We add new encoders to extract features from normals and depth. With the right choice of depth and normal [11], there is a small improvement, suggesting there may be enough information in persistent and transient maps alone to estimate lighting effects. **Note:** all qualitative comparisons in the paper are to our completely self-supervised Illuminator without normal or depth supervision.

Illuminator generalizes to other light editing applications. Illuminator is trained to abstract an illumination representation and apply it to a persistent map. This is enough to relight face portraits (Figure 10) without ever seeing portrait

Table 4. Ablation. Additional supervision from pretrained normals (N) and depth (D) [11] improves object relighting. However, self-supervised Illuminator is still quite competitive.

Method	Self-Sup	PSNR↑	MSE↓	LPIPS↓
IR++ [31]	✗✗	18.84	1574.84	0.033
Naive Cut-and-Paste		15.08	3648.84	0.042
Ours	✓	17.65	2087.03	**0.038**
Ours + N [54] + D [43]	✗	17.41	2227.01	**0.038**
Ours + N [11] + D [11]	✗	**17.78**	**2042.51**	**0.038**

Table 5. We report PSNR scores for all objects used in our quantitative evaluation. Our decomposition with our cross-loss consistently outperforms for eight out of twelve objects we studied. We observe this strong performance as our method adds or removes gloss (Barrel in Figure 8) that is consistent with the background scene lighting. Baseline IH methods struggle to make these large appearance changes.

Object	IR++ [31]	CP	Dovenet [8]	Rainnet [36]	Ours baseline	Ours (Eq. 5)	Ours (Eq. 6)
Arm Chair	18.22	14.3	14.35	14.88	15.05	16.20	**16.76**
Barrel	21.46	17.16	16.42	16.71	18.82	19.59	**20.34**
Ceramic Vase	16.24	12.78	**15.64**	13.97	13.69	15.37	15.58
Baseball	14.95	12.18	**14.61**	13.82	12.99	14.55	14.46
Lego	18.45	14.56	16.18	15.76	15.65	16.30	**16.86**
Marble Bust	15.69	12.69	**15.56**	14.29	13.64	15.24	15.49
Materials	16.67	14.65	**17.81**	16.58	15.89	16.40	17.23
Potted Plant	19.39	15.32	17.60	16.97	16.43	17.58	**17.81**
Side Table	18.99	14.67	16.11	16.00	16.14	17.34	**17.69**
Wooden Table	21.02	16.69	18.48	17.89	18.25	18.96	**19.21**
Vintage Cabinet	24.26	19.58	18.94	20.06	20.81	19.64	**21.32**
Wine Barrel	20.72	16.42	17.63	17.83	17.54	18.75	**19.11**
Mean	18.84	15.08	16.61	16.23	16.24	17.16	**17.65**

Fig. 9. Rendering in realistic hard scenarios. We transfer a set of 16 different materials rendered in a low illumination environment to bright indoor scenes with shaft lighting. Dovenet [8], a supervised SOTA IH baseline, further darkens materials. Ours generates plausibly realistic rendering by enhancing the glossiness for regions consistent with the background scene.

Fig. 10. Applications: portrait relighting. We add a new light source to our additive transient map and show acceptable portrait relights without any additional training or supervision. We also show naive composition ($P \times T_M + T_A$) for comparison.

Fig. 11. Applications: scene relighting. Illuminator can generate plausible relighted versions of the scenes by simply editing light sources. Our decomposition allows us to manipulate light sources by adding or removing them in our additive transient map. On the left, notice glossy patches are removed on the floor and pillow. On the right, a realistic strong glossy reflectance on the metallic surface of the cellphone.

relights. It can also adjust scene lighting to generate acceptable relights (Figure 3 and Figure 11), seeing no paired relighting data when training Illuminator.

9 Discussion and Conclusion

Our method, Illuminator, is completely self-supervised and trained from in-the-wild image data and does not require any labeled images for training. It can abstract illumination of the scene because of our carefully designed cross-illumination cycle consistency loss. The key to Illuminator's success is our novel image decomposition strategy into their persistent and transient maps and preserving high-frequency details in the persistent map.

Better WHDR score $\neq$ better PSNR for object relighting, likely because preserving small details causes WHDR to fall; which suggests a need to evaluate intrinsic image methods in the context of a practical task.

Limitations and future work. Our method currently cannot cast shadows. Casting shadows only from 2D image information is extremely challenging. There are supervised methods to do so for outdoor scenes [21, 37, 52] but they do not generalize well to indoor scenes (examples in supplementary). Furthermore, these methods do not correct inserted fragment's lighting as Illuminator does.

Future work will treat incorporating insights from these methods and other methods that are used to remove shadows [55]. We showed only one example of pragmatic persistent and transient decomposition. Future work will explore other decompositions (likely in feature space rather than RGB space) that meet persistent and transient criteria. Finally, we believe there must be some underlying geometric representation lurking inside the Illuminator that enables acceptable relight when using our decomposition. Future work will explore how to extract and exploit this representation for single image scene relighting and targeted scene relighting [13].

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